

Case Study 1

Employee Salaries A company records the annual salaries (in ₹ lakhs) of 9 employees:

4, 5, 5, 6, 7, 8, 8, 9, 30

1. Find the mean, median, and mode.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Identify the outlier.
5. Which measure of central tendency best represents the typical salary? Explain why

Answer:

Given	data	(in	₹	lakhs):				
4,	5,	5,	6,	7,	8,	8,	9,	30
Number	of	employees	(n)	=	9			

1. Mean, Median, and Mode

Mean

$$\text{Mean} = \frac{4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30}{9} = \frac{82}{9} \approx 9.11$$

Median

First, sort the data (already sorted).
Since $n=9$ (odd), the median is the middle value (5th value).
Order: 4, 5, 5, 6, 7, 8, 8, 9, 30

Median=7

Mode

The most frequent value is 5 (appears twice) and 8 (appears twice).
So, the data is **bimodal**:

Mode=5 and 8

2. Range

$$\text{Range} = \text{Maximum} - \text{Minimum} = 30 - 4 = 26$$

3. Variance and Standard Deviation

We use the **sample variance** (since this is a sample of employees, not the entire population).

Step 1: Mean ($\bar{x}$) = 9.11 (approx)

Step 2: Deviations from mean, squared:

Salary (x)	$x - \bar{x}$	$(x - \bar{x})^2$
4	-5.11	26.11
5	-4.11	16.89
5	-4.11	16.89
6	-3.11	9.67
7	-2.11	4.45
8	-1.11	1.23
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.39

Sum of squared deviations =
 $26.11 + 16.89 + 16.89 + 9.67 + 4.45 + 1.23 + 1.23 + 0.01 + 436.39 = \mathbf{512.87}$

Sample Variance (s^2)

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} = \frac{512.87}{8} = 64.11 \quad s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} = \frac{512.87}{8} = 64.11$$

Sample Standard Deviation (s)

$$s = \sqrt{64.11} \approx 8.01 \quad s = \sqrt{64.11} \approx 8.01$$

4. Identify the Outlier

Using the **1.5 × IQR rule**:

- Sorted data: 4, 5, 5, 6, 7, 8, 8, 9, 30
- Q1 (25th percentile) = median of lower half (4, 5, 5, 6) → $(5+5)/2 = 5$

- $Q3$ (75th percentile) = median of upper half (8,8,9,30) $\rightarrow (8+9)/2 = 8.5$
- $IQR = 8.5 - 5 = 3.5$
- Lower bound = $Q1 - 1.5 \times IQR = 5 - 5.25 = -0.25$
- Upper bound = $Q3 + 1.5 \times IQR = 8.5 + 5.25 = 13.75$

Any value < -0.25 or > 13.75 is an outlier.
30 is greater than 13.75, so:

Outlier=30

5. Best Measure of Central Tendency

Answer: The **median (7)** best represents the typical salary.

Reason:

The mean (9.11) is pulled upward by the extreme outlier (30) and does not reflect the salary of most employees. The mode (5 and 8) only tells us the most frequent values but ignores the overall distribution. The median (7) is resistant to outliers and shows that half the employees earn less than 7 and half earn more than 7, making it the most accurate representative of a “typical” salary in this skewed dataset.

Summary Table

Measure	Value
Mean	9.11 ₹ lakhs
Median	7 ₹ lakhs
Mode	5 & 8 ₹ lakhs
Range	26 ₹ lakhs
Sample Variance	64.11
Sample Std. Dev.	8.01
Outlier	30 ₹ lakhs
Best Measure	Median

Case Study 2

Website Response Time A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the mean and median.

2. Find the range.
3. Calculate the variance and standard deviation.
4. Determine whether the data is likely to be right-skewed or left-skewed.
5. Explain why the median may be more useful than the mean.

Answer:

Given data (in milliseconds):

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Number of observations (n) = 10

1. Mean and Median

Mean

$$\begin{aligned}\text{Mean} &= \frac{120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300}{10} \\ &= \frac{1505}{10} = 150.5 \text{ ms}\end{aligned}$$

Median

Since n (even), the median is the average of the 5th and 6th values.

Sorted data: 120, 125, 128, 130, **132, 135**, 140, 145, 150, 300

$$\text{Median} = \frac{132 + 135}{2} = \frac{267}{2} = 133.5 \text{ ms}$$

2. Range

$$\text{Range} = \text{Maximum} - \text{Minimum} = 300 - 120 = 180 \text{ ms}$$

3. Variance and Standard Deviation

We use **sample variance** (since this is a sample of 10 requests).

Step 1: Mean ($\bar{x}$) = 150.5 ms

Step 2: Deviations from mean, squared:

Response Time (x)	$x - \bar{x}$	$(x - \bar{x})^2$
120	-30.5	930.25
125	-25.5	650.25
128	-22.5	506.25
130	-20.5	420.25
132	-18.5	342.25
135	-15.5	240.25
140	-10.5	110.25
145	-5.5	30.25
150	-0.5	0.25
300	149.5	22350.25

Sum of squared deviations =
 $930.25 + 650.25 + 506.25 + 420.25 + 342.25 + 240.25 + 110.25 + 30.25 + 0.25$
 $+ 22350.25$
= 25580.50

Sample Variance (s^2)

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} = \frac{25580.50}{9} \approx 2842.28$$

Sample Standard Deviation (s)

$$s = \sqrt{2842.28} \approx 53.31 \text{ ms}$$

4. Skewness Direction

Comparison:

- Mean = 150.5 ms
- Median = 133.5 ms

Since **Mean** > **Median**, the distribution is **right-skewed (positively skewed)**.

Visual

explanation:

The outlier (300 ms) pulls the mean to the right, while the median remains closer to the bulk of the data (which clusters between 120–150 ms). The long tail extends toward higher values.

5. Why the Median May Be More Useful Than the Mean

Answer: The median is more useful in this case because:

1. **The mean is distorted by the outlier** (300 ms). Most response times are between 120–150 ms, but the mean (150.5 ms) suggests a slower typical response than what most users actually experience.
2. **The median (133.5 ms)** better represents the "typical" user experience because it shows that 50% of requests are completed in 133.5 ms or less, unaffected by the single slow request.
3. **Practical decision-making:** If the website team wants to know the usual performance, the median gives a realistic baseline. The mean might falsely indicate the website is slower than it really is, leading to unnecessary optimizations.

In short: For skewed data with outliers, the median is a more robust measure of central tendency than the mean.

Summary Table

Measure	Value
Mean	150.5 ms
Median	133.5 ms
Range	180 ms
Sample Variance	2842.28
Sample Std. Deviation	53.31 ms
Skewness	Right-skewed
More useful measure	Median

Case Study 3

Student Performance The marks of 12 students in a Data Science test are: 42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3.
2. Calculate the IQR.
3. Use the $1.5 \times \text{IQR}$ rule to identify any outliers.
4. Find the mean and median.
5. Explain how the outlier affects the mean. C

Answer:

Given **data** **(marks** **out** **of** **100):**
42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95
Number of students (n) = 12

1. Find Q1, Q2 (Median), and Q3

First, confirm the data is sorted (it is):
42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Q2 (Median) – since $n=12$ ($n=12$ (even)), median is average of 6th and 7th values:

- 6th value = 52
- 7th value = 55

$$Q2 = \frac{52 + 55}{2} = \frac{107}{2} = 53.5$$

Q1 (First Quartile) – median of the lower half (first 6 values):
 Lower half: 42, 45, 48, 50, 52, 52
 Median of lower half = average of 3rd and 4th values:

$$Q1 = \frac{48 + 50}{2} = \frac{98}{2} = 49$$

Q3 (Third Quartile) – median of the upper half (last 6 values):
 Upper half: 55, 58, 60, 62, 65, 95
 Median of upper half = average of 3rd and 4th values:

$$Q3 = \frac{60 + 62}{2} = \frac{122}{2} = 61$$

Summary:

- **Q1 = 49**
- **Q2 = 53.5**
- **Q3 = 61**

2. Calculate the IQR (Interquartile Range)

$$IQR = Q3 - Q1 = 61 - 49 = 12$$

3. Use the $1.5 \times IQR$ Rule to Identify Outliers

Step 1: Calculate boundaries

- Lower bound = $Q1 - 1.5 \times IQR = 49 - 1.5(12) = 49 - 18 = 31$
- Upper bound = $Q3 + 1.5 \times IQR = 61 + 1.5(12) = 61 + 18 = 79$

Step 2: Identify outliers
 Any value < 31 or > 79 is an outlier.
 Checking the data: 42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

- All values except **95** fall within 31–79
- **95** is greater than 79

Outlier=95 Outlier=95

4. Find the Mean and Median

Mean

$$\text{Mean} = \frac{42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95}{12}$$

Sum = 684

$$\text{Mean} = \frac{684}{12} = 57$$

Median (already calculated in Q2) = **53.5**

5. Explain How the Outlier Affects the Mean

The outlier (95) **pulls the mean upward** significantly.

Demonstration:

- **Mean with outlier** = 57
- **Mean without outlier** (remove 95):
 Sum = 684 - 95 = 589
 New mean = 589 / 11 = **53.55**

Comparison:

Scenario	Mean
With outlier (95)	57.0
Without outlier	53.55
Median (unchanged)	53.5

Summary Table

Measure	Value
Q1	49
Q2 (Median)	53.5
Q3	61
IQR	12

Lower bound	31
Upper bound	79
Outlier(s)	95
Mean (with outlier)	57
Mean (without outlier)	53.55
Median	53.5

Case Study 4

E-Commerce Orders An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

1. Calculate the mean, median, and mode.
2. Find the variance and standard deviation.
3. Calculate the coefficient of variation if the mean and standard deviation are known.
4. What does the standard deviation tell the company?
5. Would you describe the order data as highly or moderately variable?

Given data (number of orders per day):
18, 20, 22, 22, 24, 25, 26, 28, 30, 35
Number of days (n) = 10

1. Mean, Median, and Mode

Mean

$$\text{Mean} = \frac{18 + 20 + 22 + 22 + 24 + 25 + 26 + 28 + 30 + 35}{10}$$

Sum = 250

$$\text{Mean} = \frac{250}{10} = 25 \text{ orders}$$

Median

Since $n=10$ (even), median is the average of the 5th and 6th values.

Sorted data: 18, 20, 22, 22, **24, 25**, 26, 28, 30, 35

$$\text{Median} = \frac{24 + 25}{2} = \frac{49}{2} = 24.5 \text{ orders}$$

Mode

The most frequent value is **22** (appears twice).
All other values appear once.

$$\text{Mode} = 22 \text{ orders}$$

2. Variance and Standard Deviation

We use **sample variance** (since this is a sample of 10 days).

Step 1: Mean ($\bar{x} = 25$)

Step 2: Deviations from mean, squared:

Orders (x)	$x - \bar{x}$	$(x - \bar{x})^2$
18	-7	49
20	-5	25
22	-3	9
22	-3	9
24	-1	1
25	0	0
26	1	1
28	3	9
30	5	25
35	10	100

Sum of squared deviations =
 $49 + 25 + 9 + 9 + 1 + 0 + 1 + 9 + 25 + 100 = 228$

Sample Variance (s^2)

$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1} = \frac{228}{9} \approx 25.33$$

Sample Standard Deviation (s)

$$s = \sqrt{25.33} \approx 5.03 \text{ orders}$$

3. Coefficient of Variation (CV)

The Coefficient of Variation is:

$$CV = \left(\frac{s}{\bar{x}} \right) \times 100\%$$

Given:

- Mean ($\bar{x}$) = 25
- Standard deviation (s) = 5.03

$$CV = \left(\frac{5.03}{25} \right) \times 100\% = 0.2012 \times 100\% = 20.12\%$$

4. What Does the Standard Deviation Tell the Company?

The standard deviation of **5.03 orders** tells the company:

1. **Typical deviation from average:** On a typical day, the number of orders fluctuates by about **±5 orders** from the mean of 25 orders.
2. **Predictability:** Most days (roughly 68% if normally distributed) will have orders between:
 - a. Lower range: $25 - 5.03 = \sim 20 \text{ orders}$
 - b. Upper range: $25 + 5.03 = \sim 30 \text{ orders}$

This means the company can expect daily orders to typically fall between **20–30 orders**.

3. **Inventory and staffing:** The company should prepare for daily variations of about 5 orders. They might need flexible staffing to handle days with 30+ orders (like the 35-order day) while not overstaffing on slower days (like 18 orders).
4. **Identifying unusual days:** Days with orders below ~15 or above ~35 (mean $\pm 2 \times SD$) could be investigated for special causes (e.g., promotions, technical issues).

5. Would You Describe the Order Data as Highly or Moderately Variable?

Answer: Moderately variable

Reasoning:

Criterion	Value	Interpretation
Coefficient of Variation (CV)	20.12%	
Rule of thumb:		
CV < 15%	Low variability	Data is very consistent
CV = 15–30%	Moderate variability	Reasonable fluctuations
CV > 30%	High variability	Very unpredictable

- At **20.12%**, the CV falls in the **moderate variability** range.
- The standard deviation (5.03) is about **20% of the mean** (25), which means the day-to-day variation is noticeable but not extreme.
- The range (35 - 18 = 17 orders) is about 68% of the mean, which further supports moderate variability—not too tight, but not wildly erratic either.

Summary Table

Measure	Value
Mean	25 orders
Median	24.5 orders
Mode	22 orders
Sample Variance	25.33
Sample Standard Deviation	5.03 orders
Coefficient of Variation	20.12%
Variability level	Moderate

Case Study 5

Comparing Two Data Science Models Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

Calculate the mean error for both models.

2. Calculate the variance for both models.

3. Calculate the standard deviation for both.

4. Which model has more consistent predictions?

5. Can the model with the lower mean error necessarily be considered better? Explain.

Answer:

Given data (prediction errors):

- **Model A:** 2, 3, 3, 4, 4, 5, 5, 6
- **Model B:** 1, 1, 2, 4, 5, 7, 8, 9

Number of observations for each model (n) = 8

1. Calculate the Mean Error for Both Models

Model

A:

Sum = $2 + 3 + 3 + 4 + 4 + 5 + 5 + 6 = 32$

$$\text{Mean}_A = \frac{32}{8} = 4$$

Model

B:

Sum = $1 + 1 + 2 + 4 + 5 + 7 + 8 + 9 = 37$

$$\text{Mean}_B = \frac{37}{8} = 4.625$$

2. Calculate the Variance for Both Models

We use **sample variance** ($n-1$ denominator) since these are samples of prediction errors.

Model A (Mean = 4):

Error (x_i)	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
2	-2	4

3	-1	1
3	-1	1
4	0	0
4	0	0
5	1	1
5	1	1
6	2	4

Sum of squared deviations = 4 + 1 + 1 + 0 + 0 + 1 + 1 + 4 = **12**

$$s_A^2 = \frac{12}{8-1} = \frac{12}{7} \approx 1.71$$

Model B (Mean = 4.625):

Error ($x - \bar{x}$)	$x - \bar{x}$	$(x - \bar{x})^2$
1	-3.625	13.14
1	-3.625	13.14
2	-2.625	6.89
4	-0.625	0.39
5	0.375	0.14
7	2.375	5.64
8	3.375	11.39
9	4.375	19.14

Sum of squared deviations = 13.14 + 13.14 + 6.89 + 0.39 + 0.14 + 5.64 + 11.39 + 19.14 = **69.87**

$$s_B^2 = \frac{69.87}{8-1} = \frac{69.87}{7} \approx 9.98$$

3. Calculate the Standard Deviation for Both

Model A:

$$s_A = \sqrt{1.71} \approx 1.31$$

Model B:

$$s_B = \sqrt{9.98} \approx 3.16$$

4. Which Model Has More Consistent Predictions?

Answer: Model A has more consistent predictions.

Reasoning:

Metric	Model A	Model B
Mean Error	4.00	4.625
Variance	1.71	9.98
Standard Deviation	1.31	3.16

- **Model A's standard deviation (1.31)** is much smaller than Model B's (3.16). This means Model A's errors are tightly clustered around its mean (most errors fall between ~2.7 and 5.3).
- **Model B's errors** are widely spread (ranging from 1 to 9), with a standard deviation more than twice that of Model A.
- **Visual comparison:**
 - Model A errors: 2, 3, 3, 4, 4, 5, 5, 6 (all within a narrow 4-point range)
 - Model B errors: 1, 1, 2, 4, 5, 7, 8, 9 (spread across an 8-point range)

Model A's predictions are **more reliable and reproducible** because the error magnitude varies less from prediction to prediction.

5. Can the Model with the Lower Mean Error Necessarily Be Considered Better? Explain

Answer: No, the model with the lower mean error cannot necessarily be considered better.

Explanation with reasons:

1. **Mean error alone is insufficient:** Model A has a lower mean error (4.0 vs. 4.625), but this only tells us the *average* mistake. It doesn't tell us how much the errors fluctuate.

Final Comparison Table

Metric	Model A	Model B	Better Model
Mean Error	4.00	4.625	Model A
Variance	1.71	9.98	Model A
Standard Deviation	1.31	3.16	Model A
Consistency	High	Low	Model A
Worst Error	6	9	Model A
Best Error	2	1	Model B